

MultiVerse Anime Bot

Complete Game Design Document

2 Anime Worlds	60+ Characters	5 Rarity Tiers	8+ Mission Types
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Version	v1.0 — Initial Release
Platform	Telegram Bot (Python)
Anime worlds	Naruto, Attack on Titan (expandable)
Tech stack	Python · MongoDB · Redis · Cloudinary

1. Concept & Core Pillars

What this bot is: Players pick an anime world — Naruto or Attack on Titan — and live inside it. They collect characters, run missions, fight PvP, and grind ranks. A shared World Lobby connects both universes for cross-anime trading and epic events. Everything runs inside Telegram with inline buttons — no typing game commands, everything is tappable on mobile.

Grind that matters	Every mission, every PvP gives XP, coins, and rare drops. Players always feel progress.
Gacha with pity	Pull characters like Naruto or Levi. After 50 pulls without a Legendary, the next is GUARANTEED Legendary.

Events keep it fresh	Monthly events, seasonal raids, limited banners. Always something new for players to chase.
Mobile-first UI	Everything is a tappable button. Clean, dark, no clutter. Max 3 buttons per row. Messages under 200 chars.

2. Tech Stack

Python 3.11	Core language for all bot logic
python-telegram-bot v21	Telegram API wrapper — handlers, inline buttons, webhooks
MongoDB Atlas	Main database — users, characters, missions, events (free tier: 512MB)
Redis	Cooldowns, rate limits, session state, pity counters cache
Cloudinary / ImgBB	Character art image hosting — send via bot.send_photo(photo=url)
APScheduler	Daily resets, weekly PvP refresh, event timers, streak resets at midnight UTC
Railway / Render	Free-tier deployment. Railway recommended for always-on uptime

3. Anime Worlds

3.1 Naruto — The Hidden Leaf Village

Lore: A ninja world divided into Hidden Villages. Ninja use chakra (spiritual energy) for jutsu (techniques). Players join as a Genin (rookie) and rise through ranks. The Hidden Leaf (Konoha) is home base. Enemies: Akatsuki members, rogue ninja, and legendary villains like Orochimaru and Pain.
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Currency	Details
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Ryo	Standard ninja pay. Shop: tools, scrolls, antidotes, gear upgrades
Chakra Crystals	Premium pull currency. Earned from S-rank missions and events
Rank	Level Range
Genin	Level 1–9 Starter rank, D/C missions only
Chunin	Level 10–24 B-rank missions unlock, trade enabled
Jonin	Level 25–49 A-rank missions, World Lobby access at 10+
Anbu	Level 50–74 Special Anbu-only missions
Kage	Level 75+ Endgame rank, exclusive Kage banner
Missions	Description
D-rank: Village Errands	Escort, odd jobs in Konoha. Low risk, low reward. No cooldown.
C-rank: Patrol	Outside village walls. Encounter road bandits. 15 min cooldown.
B-rank: Capture	Chase rogue ninja. Medium risk. 30 min cooldown.
A-rank: Akatsuki	Battle Akatsuki scouts. High risk, great drops. 1hr cooldown.
S-rank: Villain Boss	Face Orochimaru, Pain, Madara. Hardest. 3hr cooldown. Best rewards.

3.2 Attack on Titan — Inside the Walls

Lore: Humanity lives inside three concentric walls — Maria (outermost), Rose, Sina — to survive Titans: massive humanoid monsters. Players join the Survey Corps, the bravest division that ventures outside the walls. ODM (Omni-directional Mobility) gear lets soldiers fly through the air and slice Titan napes. Enemies: Pure Titans, Abnormal Titans, Armored/Colossal/Female/Beast Titan.

Currency	Details
Maria Gold	Standard soldier pay. Survey Corps depot: gear, gas canisters, blades

Expedition Medals	Premium pull currency. Earned from Titan raids and Colossal events
Rank	Level Range
Cadet	Level 1–9 Training grounds, wall patrol only
Scout	Level 10–24 Venture outside Wall Rose
Elite	Level 25–49 Titan Hunt missions unlock
Captain	Level 50–74 Captain-level gear and formations
Commander	Level 75+ Endgame, Erwin/Levi-tier banner
Missions	Description
Patrol: Wall Scout	Walls scouting near Trost. Low risk. No cooldown.
Recon: Outside Rose	Venture beyond Wall Rose. Encounter Pure Titans. 15 min.
Titan Hunt	Target Abnormal Titans. Medium-high risk. 30 min cooldown.
Raid: Titan Forest	Dense forest expedition. Very high risk. 1hr cooldown.
Wall Breach Defense	Defend against Colossal Titan. Hardest. 3hr cooldown.

4. Core Game Mechanics

4.1 Core Game Loop

1	Choose your world On /start, player picks Naruto or AoT. Sets lobby, currency, rank system, shop. Can switch at Lv 10 via World Lobby (3-day cooldown penalty).
2	Pull characters (Gacha) "/gacha" pulls from character pool. Standard pull = 1 Chakra Crystal. 10-pull = guaranteed Rare+. Pity at 50 pulls = forced Legendary. Duplicate chars give Scraps.
3	Build your team Set team of 4 (Naruto) or 3 (AoT). Choose Sensei/Captain for passive bonus. Team Power = sum of char ATK+DEF+SPD with formation multiplier.

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Run missions
"/mission" shows available ranks. Higher rank = more XP+coins+rare drops but higher fail chance. Bot calculates result instantly. Cooldowns enforced via Redis.
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Fight PvP
"/pvp" challenges another player. Auto-battle formula — no human reaction time. Winner earns PvP Rating. Top 10 weekly = World Gem reward.
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Level up and upgrade
XP fills a bar. Leveling unlocks new mission ranks, raises stat caps. Characters level separately with Scraps. Stars = duplicate upgrades (1 dup = +20% stats).

4.2 Gacha & Pity System

Rarity	Drop Rate	Notes
Common	50%	Base characters, low stats. Used as fodder for upgrades.
Rare	30%	Solid stats. Good team fillers.
Epic	15%	Strong characters with unique abilities.
Legendary	4%	Elite characters. Pity guaranteed at pull #50.
World-class	1%	Rarest tier. Event banners only. Cannot be traded.

Pity counter is stored per player in MongoDB (field: pity_count). After 50 pulls without a Legendary, the next pull is FORCED Legendary (ignore RNG). Pity resets to 0 after the guaranteed pull. **10-pull bonus:** position 10 in every 10-pull is guaranteed Rare or higher.

4.3 Battle Formula

Team Power = SUM(char.ATK x 0.40 + char.DEF x 0.35 + char.SPD x 0.25)
Formation Bonus = +0% to +20% (Naruto: Sensei type match | AoT: tactic selection)
Effective Power = Team Power x (1 + Formation Bonus)
Win Chance = (myEffectivePower / totalCombinedPower) x 100
Random roll 1-100. If roll <= winChance --> WIN. Else --> LOSS
Floor rule: S-rank missions always have minimum 30% win chance.

5. Character Roster

5.1 Naruto Roster

Character	Rarity	ATK / DEF / SPD	Special Ability
Naruto Uzumaki	Legendary	95 / 70 / 85	Rasengan Barrage: +10% team ATK on win
Sasuke Uchiha	Legendary	98 / 65 / 90	Sharingan: +5% extra win chance
Kakashi Hatake	Legendary	92 / 80 / 88	Copy Ninja: copies enemy best stat for 1 turn
Sakura Haruno	Epic	75 / 90 / 70	Medical Ninjutsu: +15% DEF to all team members
Rock Lee	Rare	78 / 60 / 99	Eight Gates: SPD doubles for 1 mission (1hr CD)
Hinata Hyuga	Epic	68 / 78 / 80	Byakugan: Reveals enemy team power before fight
Jiraiya (Sensei)	World-class	99 / 85 / 78	Sage Mode: +25% to all team stats when Sensei
Itachi Uchiha	World-class	97 / 82 / 91	Tsukuyomi: 20% chance to auto-win any PvP
Gaara of the Sand	Epic	80 / 95 / 65	Sand Armor: Absorbs first loss in a mission chain
Neji Hyuga	Rare	82 / 75 / 77	Gentle Fist: Reduces enemy DEF by 10%

5.2 Attack on Titan Roster

Character	Rarity	ATK / DEF / SPD	Special Ability
Levi Ackerman	World-class	99 / 75 / 99	Ackerman Power: SPD+ATK both max. +30% when solo.
Eren Yeager	Legendary	95 / 80 / 75	Attack Titan: +15% ATK when mission fails (rage)
Mikasa Ackerman	Legendary	96 / 82 / 90	Blade Dance: Triple-attack pattern, +12% win chance
Armin Arlert	Epic	55 / 72 / 78	Strategist: +25% team win chance (brain power)

Hange Zoe	Rare	70 / 65 / 82	Research Bonus: +10% drop rate on all missions
Erwin Smith (Capt.)	World-class	88 / 90 / 72	Commander: +20% to all stats when set as Captain
Jean Kirstein	Rare	74 / 70 / 79	Leadership: +5% DEF per ally on team
Reiner Braun	Epic	85 / 98 / 60	Armored Titan: Blocks first hit in any raid mission

6. Grind Loop — Retention Mechanics

6.1 Daily Activities

- /daily claim — streak rewards. Day 7 = 5 World Gems. Day 30 = Limited character guaranteed.
- 3 free D/C-rank missions with no currency cost (stamina-free window).
- Daily shop refresh with discounted items (random selection).
- 1 free 10-pull every 7 consecutive login days.

6.2 Weekly Activities

- Weekly PvP tournament — top 10 ranked players get World Gems. Resets every Monday 00:00 UTC.
- Weekly co-op raid boss — team up 2-3 players, fight tougher boss for bigger rewards.
- Weekly challenge: specific mission type (e.g., "complete 5 B-rank missions") with bonus XP reward.

6.3 Upgrade Loops (Long-term goals)

- Character Stars: 3 duplicates raise a character from 1-star to 2-star. Max 5-star. Stats grow 20% per star.
- Gear / Equipment: craft or buy from shop. Equippable to characters for permanent stat boosts.
- Rank progression: leveling your account rank gives permanent stat bonuses to ALL characters.
- World Lobby: endgame content. Cross-anime PvP, World Gem gacha with rarest characters only.

6.4 Anti-Burnout Mechanics

- Stamina cap of 100 points. Regenerates 5 per 30 minutes. Players can binge then come back.
- Offline progress: while offline, small XP trickle from "Village Defense / Wall Guard" auto-accumulates.

- Monthly PvP leaderboard reset — everyone gets a fair shot at climbing each month.
- Pity system on gacha — players never feel cheated after many pulls without good rewards.

7. UI Design Rules (Mobile-First)

Everything must be tappable on a phone. No typing game commands. All actions use Telegram inline buttons (InlineKeyboardMarkup). Character images are sent as actual Telegram photo messages — not emojis. Messages stay short and scannable.

Rule	Specification
Button rows	Max 3 inline buttons per row. Each button label max 15 characters. Use emoji prefix (1 per button).
Message length	Bot messages under 200 characters for action screens. Profile/inventory can be longer.
Emoji usage	1 emoji per section header only. No emoji spam. Used as visual anchors, not decoration.
Stat display	Numbers + short bar representation. e.g., "ATK: 95 [=====]" — not long descriptions.
Character art	Send via bot.send_photo(photo=cloudinary_url, caption=stat_card_text). Never use image URLs in text.
Navigation	Every sub-menu has a "Back" button as last button in the last row. No dead ends.
Error messages	Short, friendly: "Cooldown! Try again in 28 min." Not technical error codes.
Loading state	For slow operations, edit the message to "Processing..." first, then update with result.

Profile card format (example)

```
[ Your Username ] -- Scout Rank
Level: 18 | XP: 4200 / 5000 [=====> ] 84%
PvP: 18 Wins / 11 Losses (62% win rate)
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Maria Gold: 1,850 | World Gems: 12
```

Team Power: 1,840 | Stamina: 87/100

Active Team: Levi [3-star] Eren [2-star] Mikasa [1-star]

8. Events Calendar

Events are the single biggest retention mechanic. They give players a deadline and exclusive rewards. Plan at least 1 event per month. Events run on APScheduler with MongoDB tracking player progress.

Chunin Exams Monthly	Bracket PvP tournament — Naruto lobby. Finals: Legendary shard + Chakra Crystals. Players sign up via /event. Auto-bracket system.
Wall Breach Defense Monthly	Server-wide co-op raid. All AoT players damage a shared Colossal Titan HP bar. If killed before timer: everyone gets Expedition Medals.
Multiverse Collision Quarterly	Cross-anime mega event. Naruto vs AoT. Special cross-anime missions unlock. Limited World-class character banner exclusive to this event only.
New Anime Launch Week On release	When new world added (e.g., Dragon Ball). 7 days of 3x XP, free 10-pulls daily, debut character banner. Massive FOMO driver.
Daily Login Streak Always on	Passive retention. Day 7 = World Gem. Day 30 = limited character. Streak breaks on miss. Highest motivation to open bot daily.
Akatsuki Hunt / Survey Expedition Seasonal (quarterly)	14-day story arc. Chapter-style missions unlock daily. Each chapter gives exclusive item. Day 14 boss fight = unique drops not available anywhere else.
Anniversary Event Yearly	1-year mark. Free 30 pulls, all-time best banner, old limited characters return. Major nostalgia + community celebration event.

9. Bot Commands Reference

Command	Description
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/start	Register or return. New players: world selection flow. Returning: profile dashboard.
/profile	Level, rank, XP bar, win rate, currencies, team power, active team display.
/team	View/edit active team. Add/remove chars, set Sensei/Captain, choose formation.
/mission	Pick mission rank. Instant result calculation. Cooldown enforced per rank via Redis.
/gacha	Pull characters. Shows pity counter. Standard / 10-pull / World pull options.
/pvp	Challenge player or enter matchmaking. Auto-resolved. Shows battle summary with stats.
/inventory	All characters, gear, consumables. Filterable by rarity and world.
/shop	Buy tools, gear, scrolls. Lobby-specific. World shop accessible at Lv 10+.
/daily	Claim daily reward + current streak. Shows next milestone reward preview.
/leaderboard	Top 10 in your lobby. /leaderboard world for cross-anime global ranking.
/trade	Send/receive character trade offers. Escrow-safe. Same-lobby validation enforced.
/coop	Create or join co-op room. 2-3 players fight a raid boss together.
/event	Current active events, your progress, rewards, and countdown timer.
/worldlobby	Access World Lobby (Lv 10+). Cross-anime PvP and World Gem gacha.
/help	Command list filtered by your current lobby. Tips for beginners.

10. Development Roadmap

Phase 1 — Foundation

Weeks 1–2

- > Set up python-telegram-bot v21 with webhook on Railway
- > MongoDB Atlas schema: users, characters, missions, events collections
- > Redis setup for cooldowns, rate limits, session state

- > /start, /profile, /help commands working with lobby selection
- > InlineKeyboardMarkup for all navigation — no plain text commands

Phase 2 — Core Gameplay

Weeks 3–5

- > Full character database with all rarities, stats, and special abilities
- > Gacha system with pity counter stored in MongoDB
- > Mission system: cooldowns, reward tables, fail/win calculation
- > Team builder with Sensei/Captain selection and formation bonuses
- > Character images via bot.send_photo() from Cloudinary URLs

Phase 3 — Social Features

Weeks 6–7

- > PvP matchmaking queue and auto-battle calculation
- > Leaderboard system (per-lobby + world global)
- > Trade system with MongoDB escrow and same-lobby validation
- > Co-op rooms: 2-3 player raid boss sessions

Phase 4 — Events & Polish

Weeks 8–10

- > Daily reward system with 7-day and 30-day streak tracking
- > World Lobby with Lv 10+ gate and cross-anime features
- > Event system with APScheduler for timers and daily resets
- > Shop system with lobby-specific item catalogues
- > Anti-spam rate limiting, anti-cheat validation, error handling

Phase 5 — Launch & Expansion

Month 3+

- > Launch with Naruto + AoT. Monitor MongoDB for performance.
- > Add Dragon Ball Z world (3rd lobby) — new chars, currency, missions
- > Add One Piece world (4th lobby) — Devil Fruits mechanic

- > Monthly character updates and limited event banners
- > Quarterly Multiverse Collision mega-event

Key advice: Build one phase at a time. A working Phase 1 with clean UI beats a broken Phase 5. Test every feature with real Telegram accounts before moving to the next phase. For character images: use AI-generated art in anime style or commission an artist. Cloudinary free tier (10GB) is plenty for launch.

11. MongoDB Schema Reference

users collection

telegram_id: Number (index)
username: String
lobby: String -- "naruto" or "aot"
level: Number -- default 1
xp: Number -- default 0
rank: String -- "Genin" | "Cadet" etc.
ryo: Number -- Naruto currency
maria_gold: Number -- AoT currency
world_gems: Number -- cross-lobby premium
pity_count: Number -- gacha pity tracker
streak: Number -- daily login streak
last_daily: Date
team: Array -- [char_id, char_id, ...]
created_at: Date

characters collection

char_id: String (unique)
owner_id: Number (telegram_id, index)
name: String -- "Naruto Uzumaki"
lobby: String
rarity: String -- Common/Rare/Epic/Legendary/World
stars: Number -- 1 to 5
atk: Number | def: Number | spd: Number
ability: String
image_url: String -- Cloudinary URL
obtained_at: Date

missions collection

user_id: Number (index)
type: String -- "B-Rank"

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status: String -- "completed"
result: String -- "win" | "loss"
rewards: Object -- {xp: 320, coins: 450}
started_at: Date
cooldown_until: Date (index)
```

pvp collection

```
p1: Number -- telegram_id
p2: Number -- telegram_id
winner: Number -- telegram_id
lobby: String
season: Number -- weekly season number
rewards: Object -- {winner_xp: 150}
timestamp: Date
```

Indexes to create: Index telegram_id on users. Index owner_id on characters. Index cooldown_until on missions (scheduler queries). Index (p1, season) and (p2, season) on pvp for weekly leaderboards. All indexes = O(1) lookups = fast bot responses.